Acid reactions

- Acid + Metal → Salt + Hydrogen

$2HCl + Mg \rightarrow MgCl_2 + H_2$

- Acid + Base → Salt + water

$H_2SO_4 + CuO \rightarrow CuSO_4 + H_2O$

- Acid + Carbonate → Salt + water + Carbon dioxide

$2HCl + Na_2CO_3 \rightarrow 2NaCl + H_2O + CO_2$

Base reactions

- Base + Acid → Salt + Water

$CuO + H_2SO_4 \rightarrow CuSO_4 + H_2O$

- Base + Ammonium Salt → Salt + Ammonia + Water

$NaOH + NH_4Cl \rightarrow NaCl + NH_3 + H_2O$

* bases are usually metal oxides / hydroxides
* soluble bases are known as alkali

acid / base effects on:

litmus

acid turns blue litmus paper to red and base turns red litmus paper blue

thymolphthalein

acid turns it colorless while base turns it blue

phenolphthalein

acid turns it colorless while base turns it pink

methyl orange

stays red-orange in acid, yellow in neutral and yellow-orange in base.

in aqueous solution...

- Lower pH → more acidic → higher concentration of H^+ ions
- higher pH → more basic → higher concentration of OH^- ions

* acids = proton donor
bases = proton acceptors

→ a strong acid is an acid that completely dissociates in aqueous solution whereas a weak acid only partially dissociates.

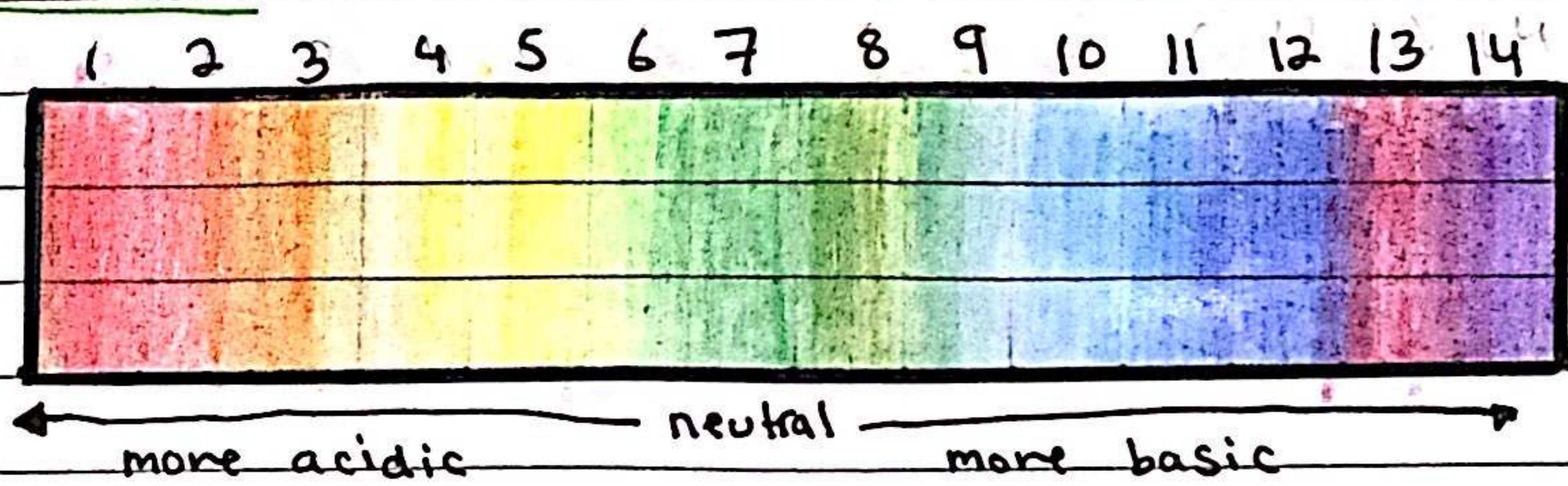
- HCl is a strong acid

$HCl(aq) \rightarrow H^+(aq) + Cl^-(aq)$

- CH_3COOH is a weak acid

$CH_3COOH(aq) \rightarrow H^+(aq) + CH_3COO^-(aq)$

pH scale



Neutralization

- reaction between an acid & base
- acid + base → salt + water

→ $H^+ + OH^- \rightarrow H_2O$

* only soluble bases (alkali) have an effect on litmus paper, insoluble bases don't

التاريخ / / | اليوم | عنوان الدرس
الموافق / / | Subject | Day Sunday Date 04/12/22

Oxide :

- a chemical compound that contains atleast 1 oxygen
- most metal oxides are basic
- most non-metal oxides are neutral or acidic
  - neutral oxides do not react with acids or bases
  - acidic oxides turn litmus paper red and neutralize bases

amphoteric oxide

- react with base and acid
- form salt + H_2O
- properties of both acids and bases

e.g Al_2O_3 & ZnO

$Al_2O_3 + 2NaOH \rightarrow 2NaAlO_2 + H_2O$ ← behaving like an acid

$Al_2O_3 + 6HCl \rightarrow 2AlCl_3 + 3H_2O$ ← behaving like a base

$ZnO + 2NaOH \rightarrow Na_2ZnO_2 + H_2O$ ← behaving like an acid

$ZnO + 2HNO_3 \rightarrow Zn(NO_3)_2 + H_2O$ ← behaving like a base

---

Salt

- a compound that is produced by the reaction of an acid & base.

preparing soluble salts ~~(titration)~~

1. use pipette to measure the alkali into a conical flask and add a few drops of indicator
2. add acid into the burette and note the starting volume
3. add the acid from the burette to the conical flask until the indicator changes to appropriate color
4. note the final volume of acid in the burette and calculate the volume of acid that was added.

⑤ add the same volume of acid into the same volume of alkali w/o the indicator
⑥ heat that solution in an evaporating basin to partially evaporate, leaving a saturated solution
⑦ Leave to crystallise

(insoluble metal/base/carbonate)
① add dilute acid into a beaker and heat it using a bunsen burner
② add the insoluble metal/base/carbonate, a tiny bit at a time, to the warm dilute acid
③ stir until the base stops disappearing
④ filture the mixture into an evaporating basin to remove excess base
⑤ heat solution so the water evaporates and the solution is saturated
⑥ Leave the [illegible] in a warm place to dry & crystallise

preparing insoluble salts (precipitation)
① dissolve soluble salts in water, and mix together using a stirring rod in a beaker
② Filter to remove precipitate from mixture
③ wash filtrate with distilled water to remove traces of other solutions
④ Leave in an oven to dry

Salts	Soluble	Insoluble
Na, K, NH_3	All	None
Nitrates	All	None
Chlorides	Most	Ag, Pb(II)
Sulfates	Most	Ba, Ca, Pb(II)
Carbonates	Na, K, NH_3	Most
Hydroxides	Na, K, NH_3, Ca (partially)	Most

Hydrated substance:
- a substance that is chemically combined to water

Anhydrous substance:
- a substance that contains no water

Water in crystallisation
- water molecules that are present in the structure of a hydrated crystal during crystallisation,

chemical formula of a hydrated compound:
* the water of crystallisation is seperated from the main formula using a dot

hydrated copper (II) sulphate
↳ $CuSO_4 \cdot 5H_2O$ ← ~~for 1 mole of $CuSO_4$, 5 moles of H_2O are~~ needed

hydrated cobalt (II) chloride
↳ $CoCl_2 \cdot 6H_2O$

* a compound which doesn't contain the water of crystallisation is called an anhydrous compound

* the conversion from a hydrated compound to a anhydrous compound is reversible, meaning the reaction can occur forwards & backwards